



8325 - Understanding the nature of the first wandering AGN candidate in a clump

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Ivan Delvecchio (PI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Giorgia Peluso (CoI) (ESA Member) (CoPI) (Contact)	INAF - Osservatorio di Astrofisica Bologna
Prof. Sirio Belli (CoI) (ESA Member)	Universita di Bologna
Ms. Rachana Bhatawdekar (CoI) (ESA Member)	ESA, European Space Astronomy Centre
Dr. Emanuele Daddi (CoI) (ESA Member)	Commissariat a l'Energie Atomique (CEA)
Dr. Maximilien Franco (CoI) (ESA Member) (US Admin CoI)	Universite Paris-Saclay
Dr. Massimo Dotti (CoI) (ESA Member)	Universita degli Studi di Milano-Bicocca
Prof. Myriam Gitti (CoI) (ESA Member)	Universita di Bologna
Dr. David Izquierdo-Villalba (CoI) (ESA Member)	University of Milano-Bicocca
Dr. Boris S. Kalita (CoI)	KIAA, Peking University
Dr. Michele Perna (CoI) (ESA Member)	Centro de Astrobiologia - CAB
Dr. Isabella Prandoni (CoI) (ESA Member)	INAF - Istituto di Radioastronomia
Dr. Alberto Sesana (CoI) (ESA Member)	University of Birmingham
Dr. Paola Severgnini (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Brera
Dr. Jasbir Singh (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Brera
Dr. Cristiana Spingola (CoI) (ESA Member)	INAF - Istituto di Radioastronomia
Dr. Alberto Traina (CoI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Dr. Francesco Ubertosi (CoI) (ESA Member)	Universita di Bologna
Prof. Gianni Zamorani (CoI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Dr. Anita Zanella (CoI) (ESA Member)	INAF - Osservatorio di Astrofisica e Scienza dello Spazio
Prof. Caitlin M. Casey (CoI)	University of California - Santa Barbara
Dr. Santosh Harish (CoI) (US Admin CoI)	Space Telescope Science Institute
Dr. Jeyhan Kartaltepe (CoI)	Rochester Institute of Technology

<i>Name</i>	<i>Institution</i>
Dr. Anton M. Koekemoer (CoI)	Space Telescope Science Institute
Dr. Daizhong Liu (CoI)	Purple Mountain Observatory, CAS
Dr. Henry Joy McCracken (CoI) (ESA Member)	CNRS, Institut d'Astrophysique de Paris
Dr. Jason D. Rhodes (CoI)	Jet Propulsion Laboratory
Prof. Brant Robertson (CoI)	University of California - Santa Cruz
Dr. Vernesa Smolcic (CoI) (ESA Member)	University of Zagreb

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(1) C2599

ABSTRACT

The growth of supermassive black holes (SMBHs) and galaxies is intertwined with the accretion history of cosmic gas reservoirs, which can fuel star formation and Active Galactic Nuclei (AGN) activity through gas filaments and/or galaxy mergers. Mergers are cosmic beacons for studying the sought-after off-center ("wandering") SMBHs, but also the formation of clumps, which are predicted to host AGN, though no such evidence has ever been reported thus far. Here we propose NIRSpec IFU observations of a unique wandering AGN candidate found in a clump, located in a merging system at $z=0.93$ already observed with NIRCам. The AGN position is inferred at $\sim 0.001''$ precision from radio VLBA data, showing $0.38''$ ($\sim 3\text{kpc}$) offset from the stellar mass peak (NIRCам/F277W) of the closest galaxy, but spot-on a clumpy structure detected in the NIRCам/F115W filter and not longward. Ancillary fiber spectroscopy sets a robust $z_{\text{spec}}=0.93$ but spatially-blended ($0.6''$ radius) among the AGN-clump and the other galaxies. Using a combined G140H/F100LP and G140M/F070LP setup, we ask for only 8.8 hours (including overheads) in order to to: (i) spectroscopically confirm the redshift of the AGN-clump structure; (ii) construct velocity maps via emission line kinematics for distinguishing between ex-situ and in-situ clump formation; (iii) obtain spatially-resolved BPT lines for interpreting gas ionization; (iv) derive reddening maps from $H\alpha/H\beta$ for investigating the puzzling dust obscuration across the system. These observations will carry an immense legacy value, opening new avenues for chasing distant wandering SMBHs and understanding their connection with galaxy mergers and clump formation.

OBSERVING DESCRIPTION

We propose NIRSpec IFU observations of the first wandering AGN candidate found in a clump, placed $\sim 3\text{ kpc}$ away from its host, and also at a record-breaking redshift of $z_{\text{spec}}=0.93$. The spectroscopic redshift is spatially-blended between the AGN-clump and two merging galaxies, over a $r=0.6''$ aperture. Hence our primary goal is to spectroscopically confirm the redshift of the AGN-clump structure. Moreover, we will study the gas

JWST Proposal 8325 (Created: Friday, February 20, 2026, 9:00:12AM Eastern Standard Time) - Overview

kinematics, gas ionization and dust obscuration across the full merging system. To achieve these goals, we plan to obtain spatially-resolved maps of the main rest-frame optical lines (H α , [NII], H β and [OIII]) with the NIRSpec/IFU. We employ two gratings/filters: G140H/F100LP for H α and [NII], which allows us to also spectrally resolve the lines and infer detailed velocity maps, and G140M/F070LP for [OIII] and H β , in order to derive BPT diagnostics and reddening maps.

The exposure time is computed with the JWST ETC to ensure an integrated $S/N \geq 15$ for the H α and [NII] lines, in each IFU (0.1"x0.1") spaxel and assuming the highest extinction over the full system ($A_v=2.5$). This conservative strategy also leads to integrated $S/N > 5-7$ at the AGN-clump position for the [OIII] and H β lines, which is high enough for BPT diagnostics.

We choose an observing strategy with 20 groups (1500 s each), and each exposure is divided into 2 integrations. We adopt the a 4-POINT-DITHER dithering pattern, and a NRSIRS2 readout pattern. Target Acquisition is unnecessary, as existing NIRCам 0.1" pointing accuracy is totally adequate for our target.

The target system can be entirely covered with only one IFU, as it is less extended than 2" across.

Background subtraction will be performed in-source using a master sky spectrum derived from empty shutters.

The total science time for our observations is 6.5 hours, with a grand total of charged time of 8.8 hours.

Proposal 8325 - Targets - Understanding the nature of the first wandering AGN candidate in a clump

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	C2599	RA: 10 01 43.4786 (150.4311608d) Dec: +02 21 34.23 (2.35951d) Equinox: J2000	Epoch of Position: 2000	
Comments: Coordinates are the (RA,Dec) center of the IFU. In-source background subtraction will be carried out using empty shutters. Therefore no companion background observation is requested. Category=Galaxy Description=[Active galactic nuclei, Active galaxies, High-redshift galaxies, Radio cores, Starburst galaxies] Extended=YES					

Proposal 8325 - Observation 1 - Understanding the nature of the first wandering AGN candidate in a clump

Observation	Proposal 8325, Observation 1												Fri Feb 20 14:00:12 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: NIRSpec IFU Spectroscopy												
	Comments: The V3 PA range was restricted to keep a bright star out of the MSA FOV and prevent spoilers in the IFU.												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	C2599	RA: 10 01 43.4786 (150.4311608d) Dec: +02 21 34.23 (2.35951d) Equinox: J2000				Epoch of Position: 2000						
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Template	TA Method						HFF Readout Mode						
	NONE						false						
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points			
	1	4-POINT-DITHER											
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Ex p	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	G140M/F070LP	NRSIRS2	20	2	false	true	NONE	4	8	11787.823	223524	
	2	G140H/F100LP	NRSIRS2	20	2	false	true	NONE	4	8	11787.823	223524	
Special Requirements	Aperture PA Range 249.06164917 to 254.21164917 Degrees (V3 110.09 to 115.24)												